

Next-Gen Indian Statutory Payroll & AI Tax Advisory System

Capstone Submission — AY 2026–27

Author

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Discipline

Full-Stack Vibe Coding & LLM Systems Engineering

Statutory Scope

Indian Income Tax Act (AY 2026–27) · EPF · ESI · Professional Tax

Tech Stack

React 19 · TypeScript · Express · Google Gen AI SDK

Date: September 25, 2026



CONTEXT & OBJECTIVES

The Indian Payroll & Tax Dilemma

Statutory Maze

Dual Old/New regimes for AY 2026–27; standard deductions of ₹75,000 vs ₹50,000; Section 87A rebate caps; EPF 12% with optional ₹15,000 wage ceiling; ESI on gross wages ≤ ₹21,000; state-specific Professional Tax; and the three-condition HRA exemption under Section 10(13A).

Employee Pain Points

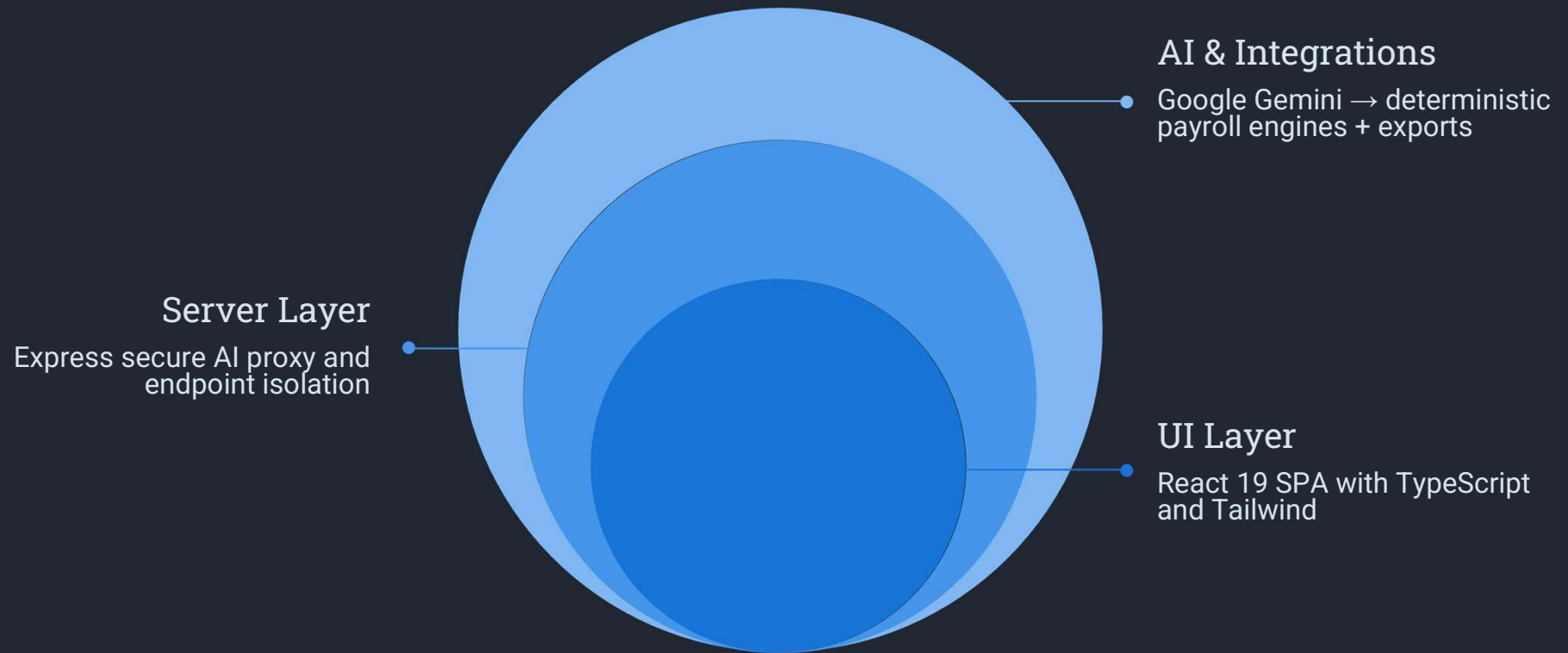
Wrong regime selection costs employees thousands. Last-minute investment-proof anxiety, opaque TDS deductions, and zero access to personalized HR guidance compound the problem year after year.

Capstone Solution

Deterministic, auditable TypeScript calculations paired with a Google Gemini AI advisor and a unified HR/employee self-service portal — transparent, explainable, and enterprise-ready.



Architecture & Full-Stack Technology



Frontend

- React 19 · strict TypeScript · Vite 6
- Tailwind CSS v4 · Lucide icons
- html2canvas · jsPDF · pptxgenjs

Backend & AI

- Node.js / Express / tsx server
- Secure endpoint isolation for API keys
- @google/genai · timeout & model fallback
- xlsx export engine

FEATURE DELIVERABLES

Core Application Modules



Master Salary Structuring

Full CTC breakdown with PF wage-ceiling toggle and component configuration.



Employee Self-Service

Payslip downloads and regime declarations through a dedicated employee portal.



Old vs New Simulator

Live side-by-side regime comparison with real-time tax differential insights.



April–March Payroll Run

Gross-to-net register with EPF/EPS, ESI, PT, and progressive TDS computation.



Declarations & HRA

Chapter VI-A (80C/80D/24b), HRA rent entry and landlord PAN validation.



Gemini AI Advisor

Smart Tax Advisor and persistent floating Copilot powered by Google Gemini.



Prototype Limitation: Comprehensive statutory edge-case testing is still in progress. The application requires full validation against the Income Tax Act, 2025 and applicable new forms/rules before any production deployment.

Deterministic Indian Tax Engine

New Regime Slabs – AY 2026–27

Income Range	Rate	Std. Deduction
₹0 – ₹3,00,000	Nil	₹75,00087A rebate ≤ ₹7L net
₹3L – ₹7L	5%	
₹7L – ₹10L	10%	
₹10L – ₹12L	15%	
₹12L – ₹15L	20%	
Above ₹15L	30%	
Income Range	Rate	
₹0 – ₹2,50,000	Nil	
₹2.5L – ₹5L	5%	
₹5L – ₹10L	20%	
Above ₹10L	30%	

₹50,000 std. deduction · 80C/80D/80CCD(1B) · 4% cess on total tax.

① Actual HRA received

② Rent paid – 10% of Basic

③ 50% Basic (metro) / 40% (non-metro)

HRA Exemption – Section 10(13A)

Exempt amount = **minimum of** three conditions:

Progressive TDS Smoothing

Monthly TDS = **(Projected Annual Tax – Tax Deducted to Date) ÷ Remaining Months**, re-computed each payroll run to prevent year-end spikes.

⊗ Legal tax and bank-deduction math must remain **deterministic** – never LLM-generated. Verify all slabs against current law before production use.

AI ADVISORY

Google Gemini Integration & Circuit-Breaker Architecture



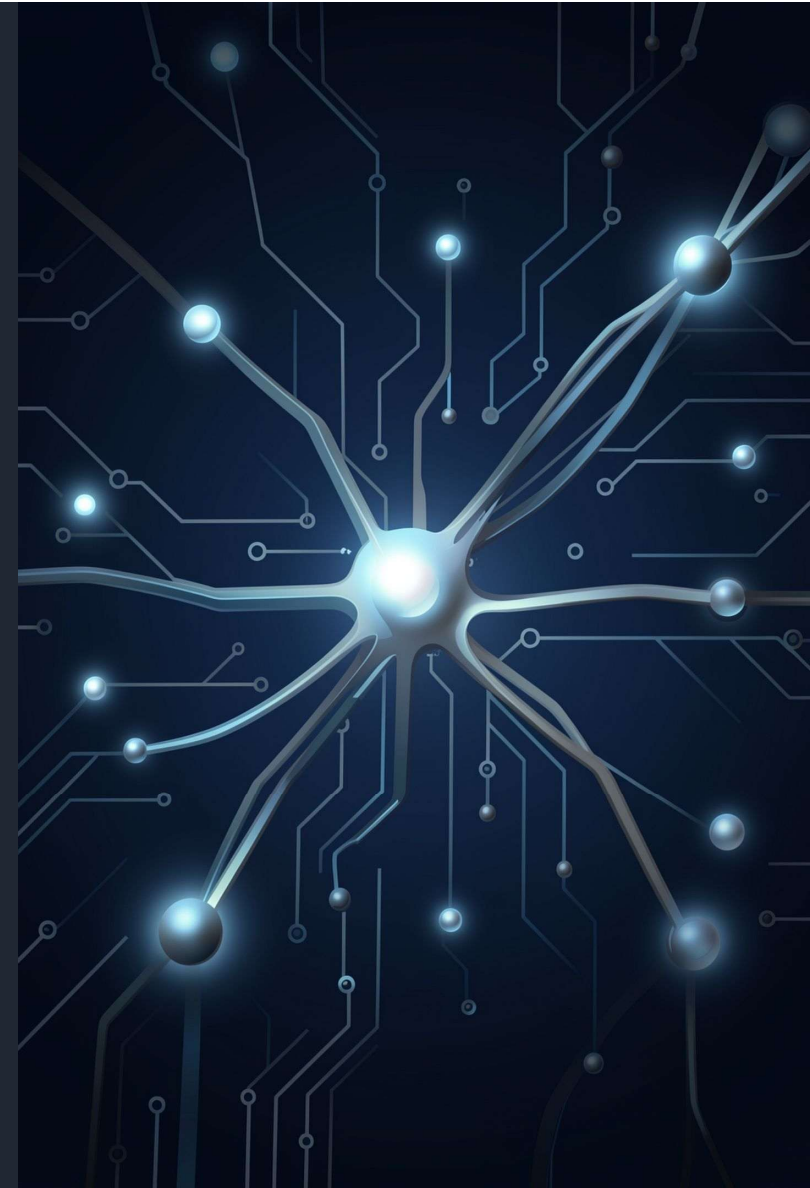
Context
Ingest

Gemini
Cascade

12s Abort &
Rules

Advisor UI &
Copilot

 **Design Principle:** AI advises and explains — deterministic TypeScript code computes all statutory liabilities. The AI layer never overwrites calculated tax figures.



METHODOLOGY

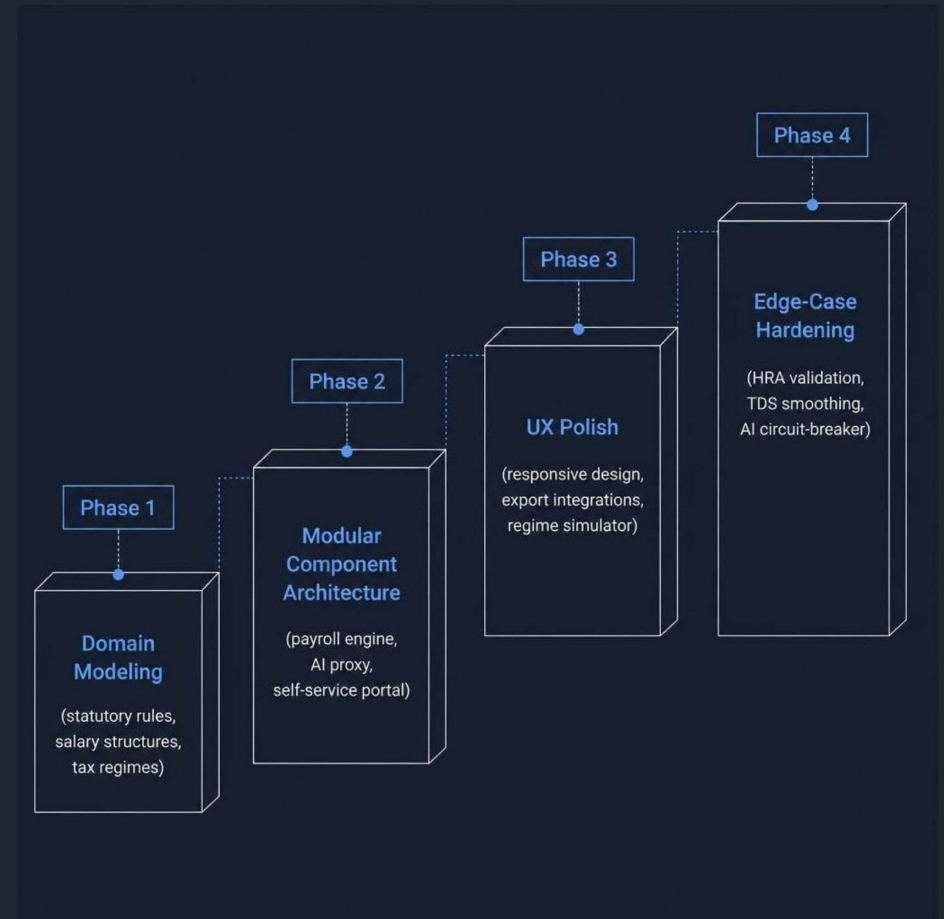
Vibe Coding: From Idea to Enterprise Prototype

What Is Vibe Coding?

High-velocity, intent-driven development where natural-language prompts and domain specifications guide AI-assisted code generation, testing, and iterative refinement — compressing weeks of scaffolding into focused engineering sprints.

Quality Controls Applied

- Strict TypeScript interfaces across all domain boundaries
- `tsc --noEmit` and Vite production builds as CI gates
- Clean separation: UI components · pure calculation utilities · server API routes



Data Security, Confidentiality & Privacy

✓ Current Strengths

API Key Isolation

GEMINI_API_KEY confined to server layer — never exposed to client bundle.

Calculation Integrity

Deterministic engine ensures statutory figures are never AI-mutated.

Graceful Degradation

Timeout-based fallback to rule-based CA guidance when Gemini is unavailable.

⚠ Prototype Gaps to Address

Session State

In-memory client-side sessions — no persistent server-side authentication.

BOLA / IDOR Risk

Employee objects accepted directly by AI routes — server-side authorization required.

PII at Rest

PAN, Aadhaar, and bank data currently unencrypted — field-level encryption needed.

Statutory Obligations

DPDP Act 2023 consent · IT Act Sec 192 · Form 24Q/Form 16 · immutable audit logs required for salary changes, regime locks, and declaration approvals.

Production Checklist & Deployment Roadmap

Pillar	Target Control	Prototype Status
Authentication	SAML/OIDC SSO · MFA enforcement	In-memory session only
Authorization	Server-side RBAC · tenant isolation	Client-side role flags
Data Persistence	Relational DB · AES-256 at rest	No persistent store
PII Protection	Field-level encryption: PAN, Aadhaar, bank	Plaintext in state
Audit Logging	Append-only immutable ledger	Not yet implemented
Infrastructure	Docker/Cloud Run · WAF · autoscaling · DDoS protection · multi-region failover	Local dev server

 All items marked  are scoped for post-prototype hardening before any production or regulated-environment deployment.

Achievements, Learnings & What's Next

Key Achievements

- Full-stack dual-regime engine for AY 2026–27
- Six core modules with HR and employee views
- Resilient Gemini advisory with CA fallback
- Excel, PDF payslip, and PPTX export pipelines

Key Learnings

- Conversational specs accelerate prototyping significantly
- Strict TypeScript interfaces prevent domain drift
- Legal math belongs in deterministic code — never AI
- Continuous compilation catches defects early

Future Roadmap

- Direct NEFT/NACH bank integration
- Form 16 Part A/B + Income Tax Act 2025 new forms
- Biometric attendance & automated LOP processing
- Cloud SQL · enterprise SSO · SOC 2 readiness
- Expanded statutory testing & edge-case validation

Explore the Prototype

GitHub Repository

github.com/joydeepdatta-lab/B89Capstone-Joydeep-Datta

Live Application

payroll16092026.ai.studio



Important: Prototype for demonstration and academic evaluation only. Statutory rules and all production security controls require final professional validation before regulated deployment.

*Submitted by **Joydeep Datta** · September 25, 2026*